

Leonard Bizarro Mokriski

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EDUCATION

UC Santa Barbara

Ph.D. in Mathematics

- Teaching Assistant
- Student Aid Fellowship

Atlanta, Georgia

September 2026 - Present

Emory University

B.S. Mathematics, B.A. Philosophy

- GPA: 3.96/4.00
- Inducted into Phi Beta Kappa Honors Society: Spring 2026
- Dean's List: Fall 2022 - Present
- Honors in Mathematics

Atlanta, Georgia

August 2022 - May 2026

Princeton University Summer Program

Summer School in Fluid Dynamics

- [NSF-FRG Collaboration: Singularities in Incompressible Flows](#)
- Subtopics: Computer Assisted Proofs and Physics-Informed Neural Networks

Princeton, New Jersey

August 2025

RESEARCH

IOL Lab - Quantum Mathematics

Entanglement detection and Koopman-von Neumann mechanics

- Research Lead: Dr. Patrick Gelß
- Invited to the [Quantum Mathematics research area](#) as a visiting researcher for summer 2026.

Zuse Institute and Technische Universität Berlin

Summer 2026

Honors Thesis in Mathematics

Schwarz-Christoffel Mappings onto Comb Domains

- Advisor: Dr. Milivoje Lukić
- Developed methods for solving the parameter problem of Schwarz-Christoffel maps onto periodic comb domains.
- Produced an original thesis paper, presented and reviewed by a thesis committee consisting of faculty from the Emory Math Department.

Emory University

Fall 2025 - Spring 2026

2025 Research Experience for Undergraduates

Shell Models for Turbulence

- Mentor: Dr. Vincent R. Martinez
- Research on toy models for fluids derived from the Navier-Stokes Equations: generalized the Sabra shell model for turbulence.
- Producing a manuscript in preparation for publication.

City University of New York

Summer 2025

Scholarly Inquiry and Research Experience

Loop Code Representation for Knots

- Selected for the Scholarly Inquiry and Research Experience (SIRE) program; awarded full funding for independent research under the supervision of Dr. Suresh Venapally.
- Investigated loop code representations of knots to analyze symmetry groups and invariants.

Emory University

Fall 2024 - Spring 2025

2024 Research Experience for Undergraduates

Characterizing Maximal Ribbon Knots

- Mentors: Dr. Jeffrey Meier and Dr. Alexander Zupan

City University of New York

Summer 2024

PRESENTATIONS

Joint Mathematics Meeting

Toy Models of Fluid Equations & Applications of Physics-Informed Neural Networks

Presentation of our upcoming paper, [Generalized Shell Models for Turbulence](#).

Washington DC

January 7, 2026

Polymath Jr. REU Conference

Generalized Shell Models for Turbulence

Virtual

August 10, 2025

Undergraduate Research Symposium

Loop Code Representations for Knots

Interviewed for my work in the [Undergraduate Research Symposium Highlights](#).

Atlanta, Georgia

April 25, 2025

WORK

Emory Math Department Student Worker

Front Desk and Office Assistant

Emory University

September 2025 - Present

- Co-organized departmental events, including the biweekly Graduate Tea and undergraduate study sessions.
- Assisted with departmental logistics for major events, including research conferences and faculty recruitment visits.
- Contributed to the department's *In Sum* magazine, authoring a [piece on EUMMA in the Fall 2025 edition](#).

Handshake AI Training

MOVE Program Fellow and Reviewer

Remote

Summer 2025 - Present

- Developed and evaluated mathematical prompts to assess the performance of Large Language Models in solving competition level mathematics problems.
- Awarded for top-tier performance and high quality contributions for expert training, and promoted from Handshake MOVE Fellow to Reviewer.

The Ben Franklin Academy High School

Teaching Assistant for Math and Physics

Atlanta, Georgia

Fall 2023 - Spring 2024

- Tutored high school students in advanced courses such as AP Calculus, AP Physics, and AP statistics.
- Constructed and graded original homework sets, quizzes, and exams for these classes.
- Assisted with class procedure and school events, including breaking down complex mathematical topics for students and promoting inclusive and accessible learning environments.

EXTRACURRICULARS

Emory University Mathematics Majors Association

Cofounder, Vice President, and Head of Academic Events

Emory University

Spring 2023 - Present

- Coordinated academic events for math majors at Emory.
- Collaborated with the Emory Math Department and faculty.
- Led talks and presentations relating to research and careers in mathematics.

Emory International Relations Association

Model United Nations Travel Team Member

Emory University

Fall 2022 - Present

- Competed in numerous Model UN conferences across the United States.
- Received awards and distinctions for performances at conferences.

Emory Philosophy Club

Researcher for Presentations and Events

Emory University

Fall 2023 - Present

- Conducted background research to prepare philosophy events and discussions.
- Constructed presentations and gave talks about various philosophical topics.

SKILLS

Python Programming:

- Libraries: NumPy, Pandas, SciPy, Scikit-learn, TensorFlow, PyTorch, JAX.
- Implemented Physics-Informed Neural Networks (PINNs), Graph Neural Networks, and Transformers for data-driven approaches to solving ODEs, PDEs, and the parameter problem for Schwarz-Christoffel maps.
- Developed numerical simulations for turbulence modelling to obtain energy spectrum estimates in fluid dynamics research.

MATLAB:

- Numerically computed Schwarz-Christoffel mappings using the [sc-toolbox](#) by Dr. Tobin A. Driscoll.
- Extended and refined the toolbox algorithms to accommodate general periodic comb domains for spectral analysis.

Julia Programming:

- Libraries: DifferentialEquations.jl, FFTW.jl.
- Computed energy estimates and spectral budgets for generalized shell models of turbulence.
- Utilized conformal mapping packages to corroborate MATLAB results and develop new methods.

Languages: English (Native), Portuguese (Fluent).

Citizenship: USA, Portugal (Dual Citizen).